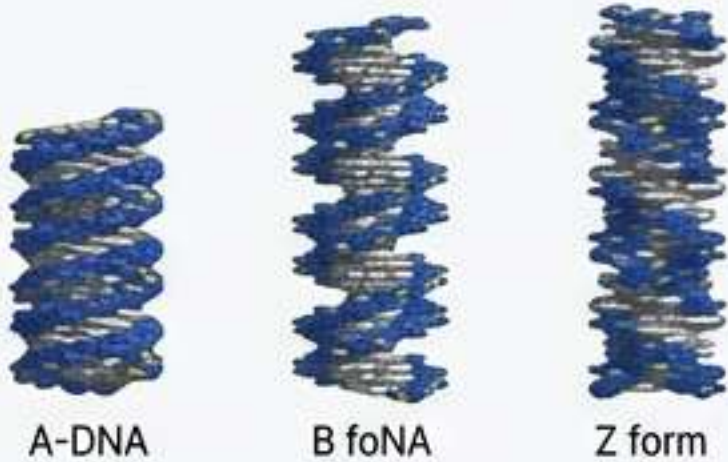


Question 1 | DNA Helix Forms

Which statement about DNA helix forms is correct?

- A. A-DNA is the most important biological form.
- B. **B-DNA is a right-handed helix with ~10 bp per turn.**
- C. Z-DNA is a left-handed mirror copy of B-DNA.
- D. Z-DNA is artificial and does not exist in nature.
- E. B-DNA is more condensed than A-DNA.



Teacher Mode: B-DNA is the standard physiological form (Watson & Crick), a right-handed helix with 10 base pairs per turn. A-DNA is dehydrated/compact. Z-DNA is left-handed and 'zigzagged', found in GC-rich regions.

Mnemonic: B = Biological (Standard).
Z = Zigzag (Left-handed).

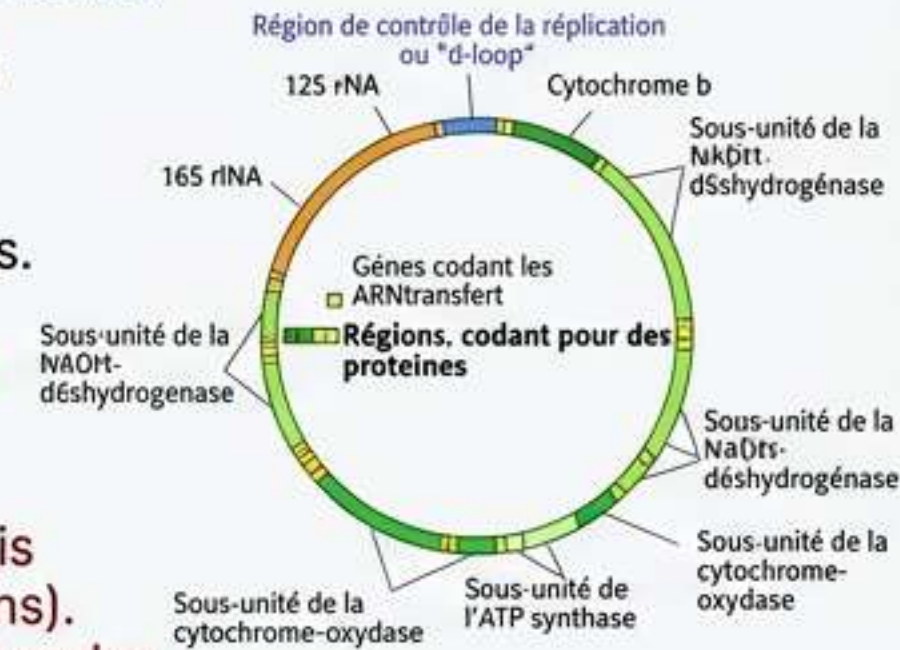
Question 2 | Mitochondrial DNA

Characteristics of Mitochondrial DNA:

- A. It is linear.
- B. Codes for mRNA but not tRNA.
- C. Codes for all mitochondrial proteins.
- D. Transmitted by both parents.
- E. **Circular, double-stranded, genes without introns.**

Teacher Mode: Like bacteria, mtDNA is **Circular** and highly efficient (No Introns). Inheritance is **exclusively Maternal**. It codes for only 13 proteins; the nucleus codes the rest.

Mnemonic: Mito Mom Circle (Maternal, Circular, No Introns).

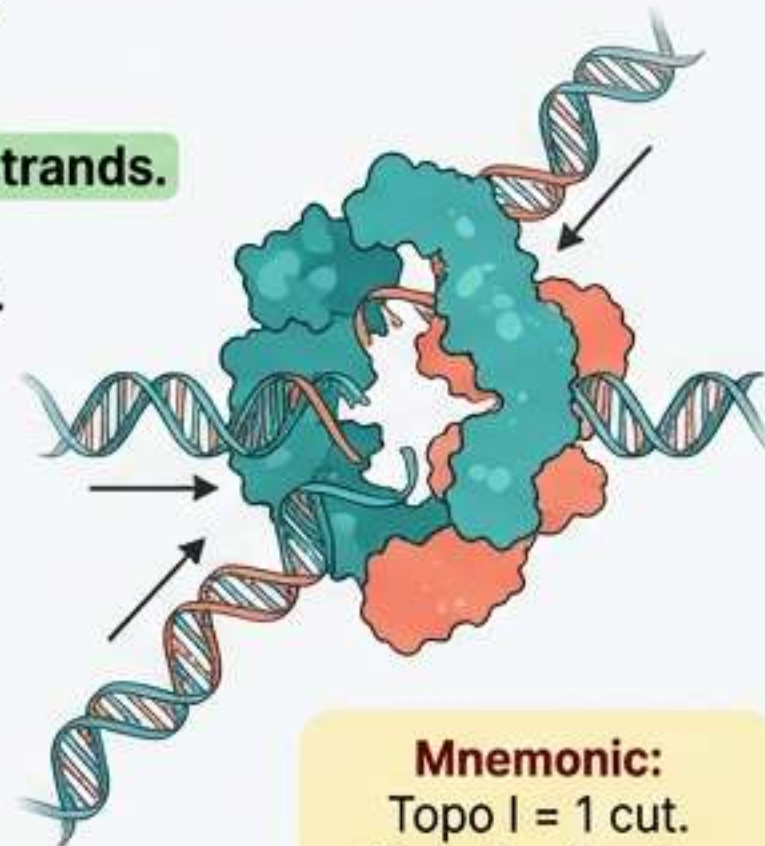


Question 3 | Topoisomerases

Role and mechanism of Topoisomerases:

- A. **Some topoisomerases cut both DNA strands.**
- B. Cannot modify DNA supercoiling.
- C. Types I and II differ by subunit number.
- D. They create knots in DNA.
- E. Topoisomerase I cuts both strands.

Teacher Mode: Topoisomerases manage supercoiling stress. **Type I** cuts one strand coritin one strand (relaxation). **Type II** (e.g., Gyrase) cuts **BOTH** strands to manage tangles.



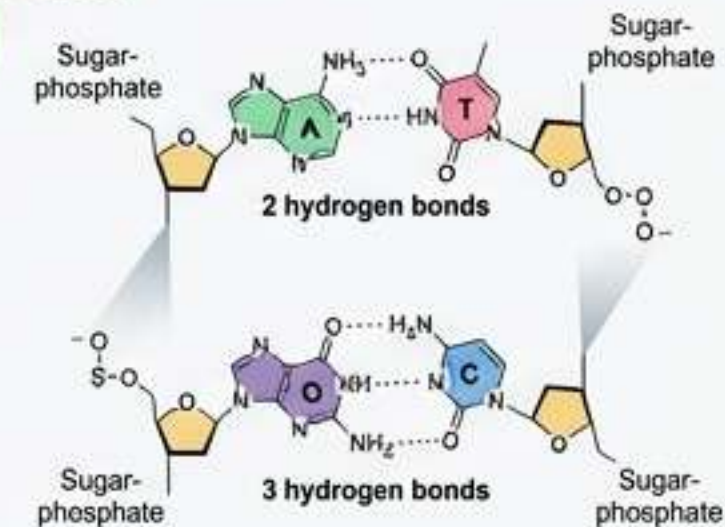
Mnemonic:
Topo I = 1 cut.
Topo II = 2 cuts.

Question 4 | Complementarity

The two DNA chains are complementary because:

- A. **Bases bind one-to-one between chains.**
- B. Sugar-phosphate interactions.
- C. Purine always matches Purine.
- D. Adenine pairs with Uracil.
- E. Cytosine pairs with Adenine.

Teacher Mode: Driven by Hydrogen bonding in the helix core. **Purines (A, G)** pair with **Pyrimidines (T, C)**. A pairs with T (2 bonds). G pairs with C (3 bonds).

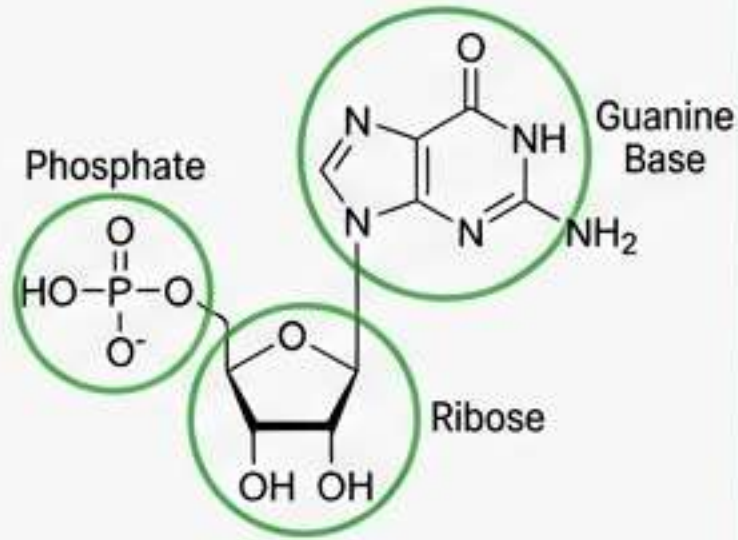


Mnemonic:
Pure As Gold (Purines).
Cut The Pie (Pyrimidines).

Question 5 | Nomenclature

Regarding nucleotide nomenclature:

- A. Adenine is a purine nucleotide.
- B. Thymidylic acid is a ribose nucleotide.
- C. Guanylate is a purine nucleotide.**
- D. Deoxycytidylic acid is an RNA nucleotide.
- E. Uridine contains a phosphate.



Teacher Mode: Suffixes matter!

Guanylate = Base + Sugar + Phosphate (Nucleotide).

Adenine is just a Base. Uridine is a Nucleoside (No Phosphate).

Mnemonic: -osine/-idine = Side (Sugar only). -ylate/-ylic = Tide (Phosphate).

Question 6 | Chargaff's Rules

In a DNA molecule:

- A. Cytosine quantity equals Thymine.
- B. Adenine quantity equals Thymine.**
- C. Thymine quantity equals Uracil.
- D. Quantities of all four bases are different.



Teacher Mode:

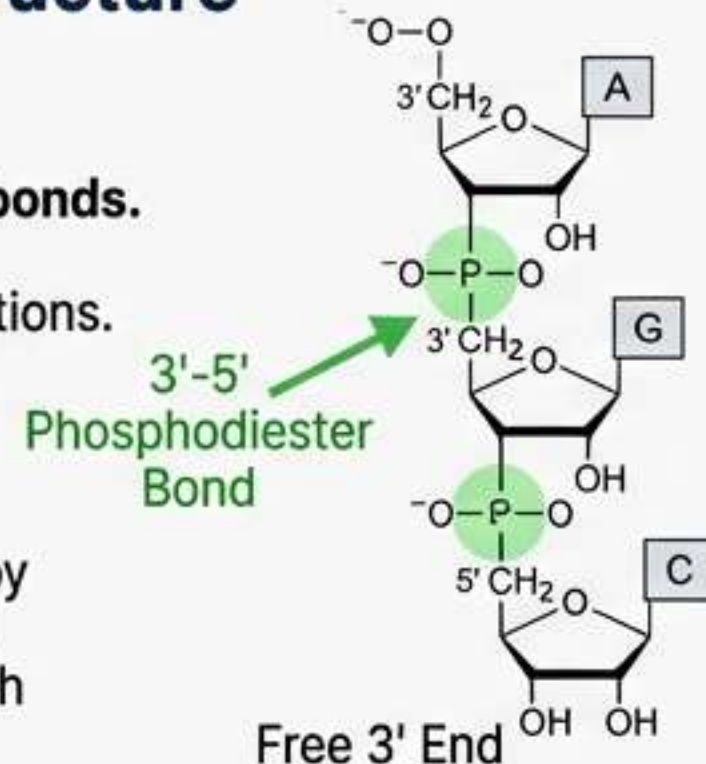
Chargaff's Rule: In dsDNA, every A pairs with T, and G with C. Therefore, molar concentration $[A] = [T]$ and $[G] = [C]$. Purines = Pyrimidines.

Mnemonic: Chargaff's Balance: A=T and G=C.

Question 7 | Backbone Structure

In nucleic acid structure:

- A. Strands have 3'→5' polarity.
- B. Nucleotides link via phosphodiester bonds.**
- C. Free 3' end contains a phosphate.
- D. Each nucleotide has two free acid functions.
- E. Base attaches to sugar via ester bond.



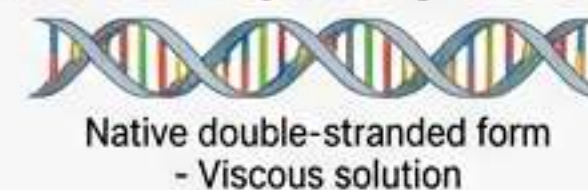
Teacher Mode: DNA is a polymer linked by **3'-5' phosphodiester bonds**. The 3' end always has a free -OH group. Bases attach via **N-Glycosidic bonds**.

Mnemonic: Phosphodiester connects the P-eople. 3' is Free (OH).

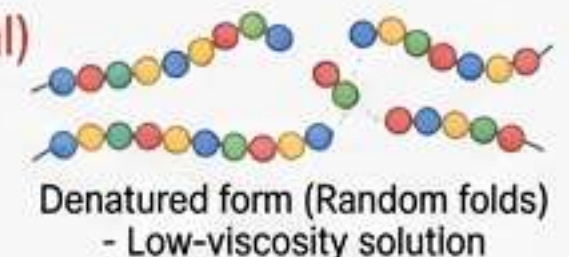
Question 8 | Denaturation

Consequences of thermal denaturation (Find the FALSE statement):

- A. Dissociation of two strands.
- B. Increased absorbance at 260nm.
- C. Decreased viscosity.
- D. Increased density.
- E. Alteration of primary structure. (Alert Coral)**



Heat



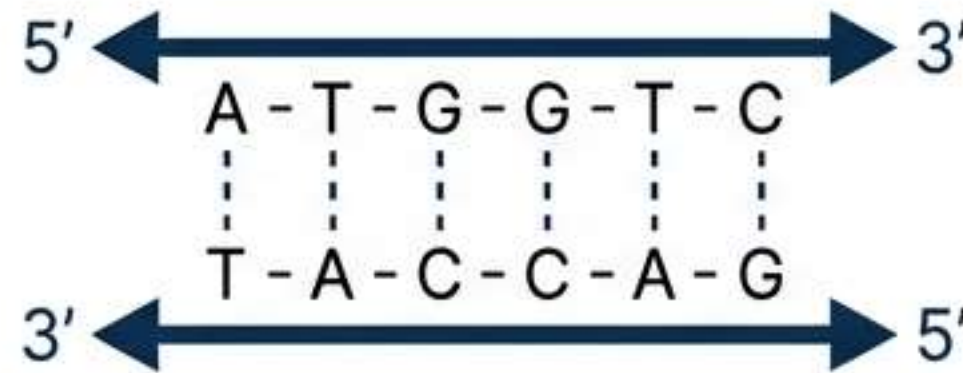
Teacher Mode: Denaturation breaks weak **Hydrogen bonds** (strand separation), but NEVER breaks the phosphodiester bonds. The **Primary Structure (Sequence)** remains intact.

Mnemonic: Heat hurts H-bonds. The Skeleton (Sequence) stays.

Question 10 | Complementary Sequence

What is the complementary strand for: 5' ATGGTC 3'?

- A. 3' TACCAG 5'
- B. 5' TACCAG 3'
- C. 5' CACCAT 3'
- D. 5' TACCAT 3'
- E. 3' TACCAT 5'



Teacher Mode: DNA is **Antiparallel**.

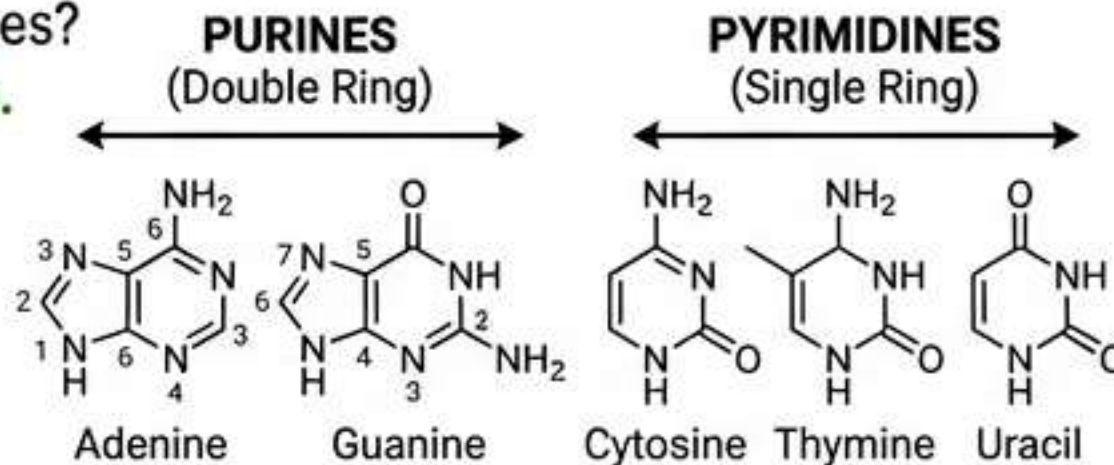
1. Flip Direction: 5'→3' becomes 3'←5'.
 2. Swap Bases: A→T, G→C.
- Result: 3' T A C C A G 5'.

Mnemonic: The 'Flip & Swap'. Always check the ENDS (5' / 3') first!

Question 14 | Purines

Which are the Purine bases?

- A. Adenine and Guanine.
- B. Cytosine.
- C. Uracil.
- D. Thymine.
- E. Pseudouridine.



Teacher Mode: Nitrogenous bases families:

Purines: Two rings (Hexagon + Pentagon). Members: Adenine & Guanine.

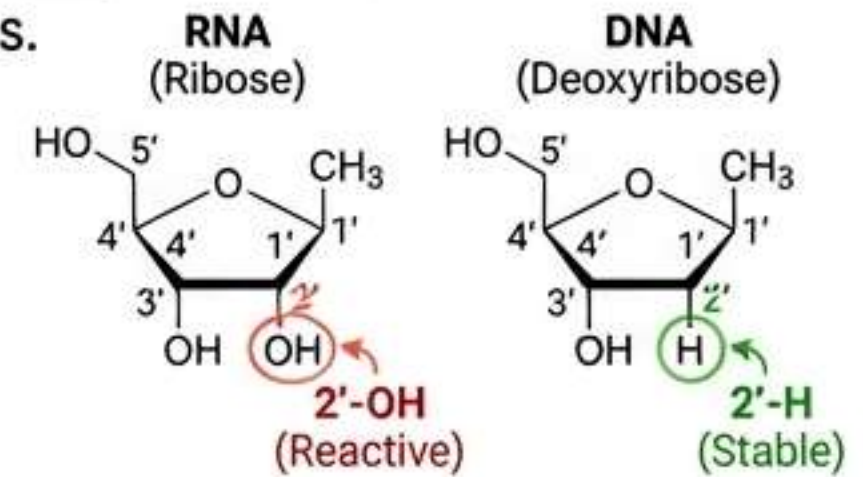
Pyrimidines: One ring. Members: Cytosine, Thymine, Uracil.

Mnemonic: PUre As Gold (Purines = A & G).

Question 12 | DNA vs RNA

Nucleic Acid properties (Find the **FALSE** statement):

- A. Formed by phosphodiester bonds.
- B. Support genetic information.
- C. Control protein synthesis.
- D. Two classes: DNA and RNA.
- E. dsDNA and RNA are both hydrolyzed by NaOH.



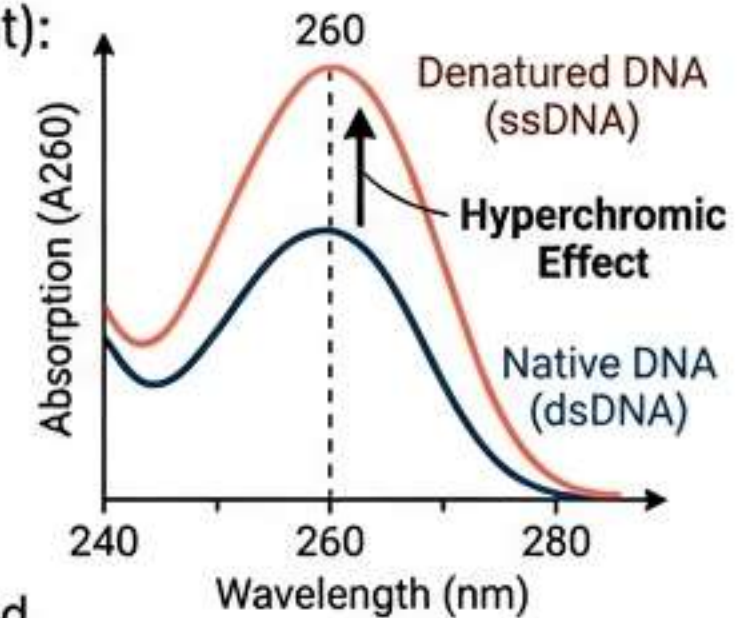
Teacher Mode: RNA has a 2'-OH group that attacks the backbone in alkaline (NaOH) conditions, destroying it. DNA has a 2'-H, making it resistant to base hydrolysis.

Mnemonic: DNA is Durable. RNA is Reactive (destroyed by Base).

Question 15 | UV Absorption

DNA Properties (Find the **FALSE** statement):

- A. Max absorption at 260nm.
- B. Single-strand absorbs more than double-strand.
- C. Denaturation breaks hydrogen bonds.
- D. T_m is temp where 50% is denatured.
- E. DNA absorbs significantly MORE than free bases.



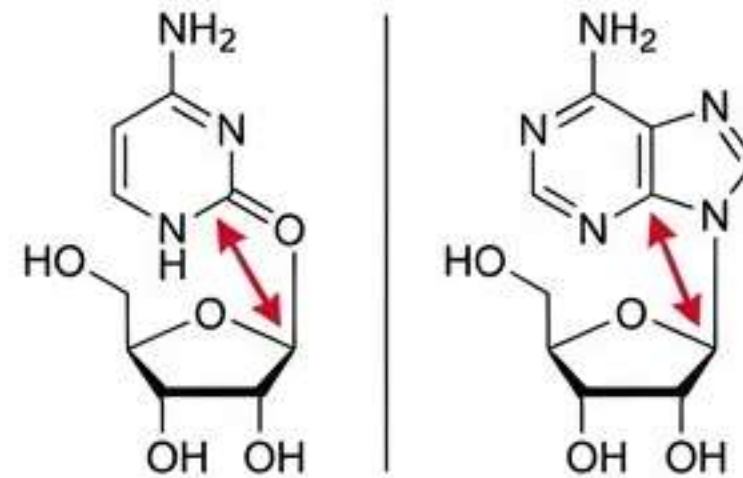
Teacher Mode: Hypochromic Effect: Stacked bases in the double helix 'hide' from UV light. Intact DNA absorbs **LESS** than free bases. Denaturation increases absorption.

Mnemonic: Stacked = Shaded (Double Helix absorbs LESS).

Question 16 | Glycosidic Bonds

A nucleoside is composed of:

- A. Pyrimidine linked to pentose via N1-C1'.**
- B. Purine linked via N1-C1'.
- C. Pyrimidine linked via N9-C1'.
- D. Purine linked via N9-C5'.
- E. Pyrimidine linked via N1-C5'.



Teacher Mode: The N-Glycosidic bond connects Base to **C1'** of sugar.

Pyrimidines (Small Ring): Connect via **N1**.

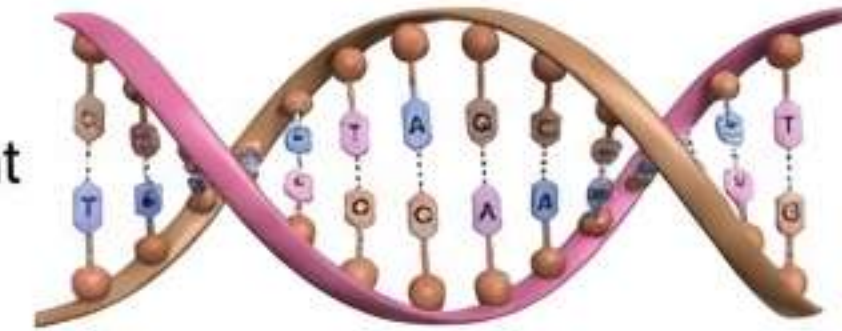
Purines (Big Ring): Connect via **N9**.

Mnemonic: Pyrimidines = Small Name (N1). Purines = Big Name (N9).

Question 18 | Watson-Crick Model

Regarding DNA structure:

- A. Bicatenary molecule of two chains.**
- B. Right helix, 10.5 bp/turn, covalent bonds between strands.
- C. Left helix, 10.5 bp/turn.



Teacher Mode: DNA is bicatenary (double-stranded).

Option B is wrong: Strands are linked by **Hydrogen bonds**, not covalent.

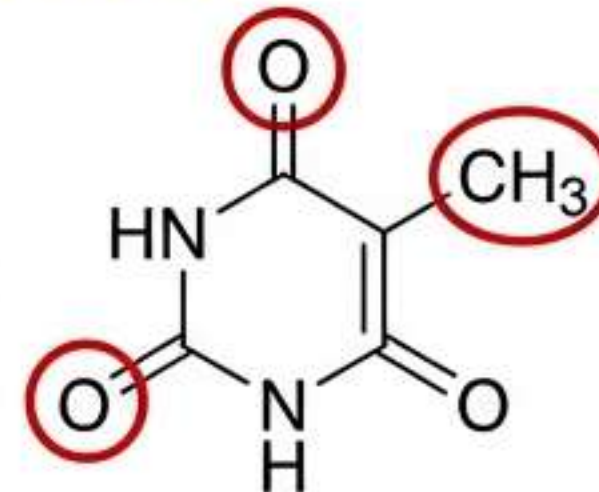
Option C is wrong: Watson-Crick DNA is a **Right**-handed helix.

Mnemonic: Bicatenary = Bi (Two) + Chain.

Question 25 | Base Structure

Nitrogenous Base Structure:

- A. Guanine has 4 Nitrogen atoms.
- B. Uracil is methylated Thymine.
- C. Thymine possesses two C=O functions.**
- D. Cytosine has a C=O on C4.



Teacher Mode:

Thymine features: Two Keto (C=O) groups (at C2 and C4) AND a **Methyl** group (at C5).

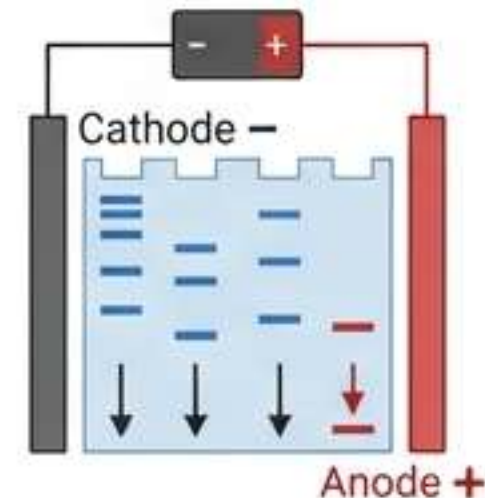
Correction: Uracil is *demethylated* Thymine. Guanine has 5 Nitrogens.

Mnemonic: Thymine has Two Tones (C=O) and a Tail (Methyl).

Question 31 | Physical Properties

Physico-chemical properties:

- A. DNA settles at bottom in centrifugation.
- B. ssDNA is more viscous than dsDNA.
- C. Positive charge at physiological pH.
- D. Acidic character; migrates to Anode (+).**
- E. Soluble in alcohol.



Teacher Mode:

DNA is an **Acid** (Deoxyribonucleic Acid). Phosphate groups are negative at pH 7. In electrophoresis, Negatives migrate to the **Positive Anode**. DNA precipitates in alcohol.

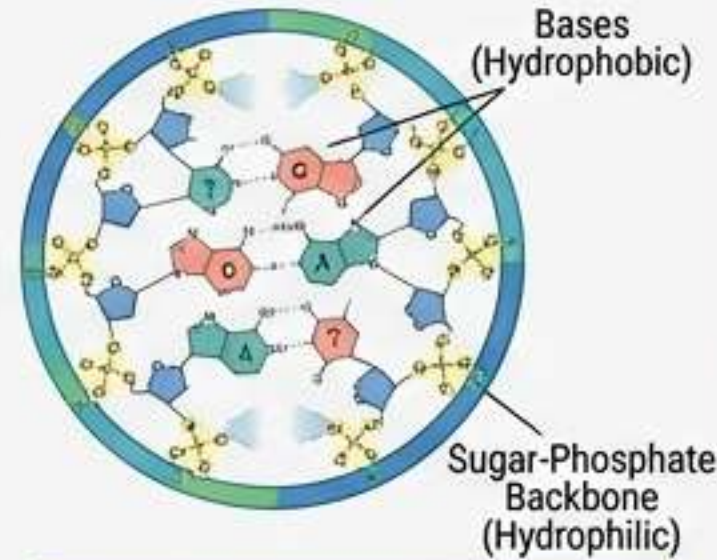
Mnemonic: Run to Red. (DNA runs to the + Anode)

Question 38 | DNA Hydrophobicity

Regarding DNA Structure (Find the FALSE statement):

- A. Molecules formed of 2 antiparallel chains.
- B. Nitrogenous bases face the exterior.**
- C. Phosphate forms 3'-5' bridge.
- D. Deoxyribose and Phosphates are on the outside.

Teacher Mode: Structure is driven by hydrophobicity. Bases are **hydrophobic**, so they **hide INSIDE**. The Sugar-Phosphate backbone is **hydrophilic**, so it faces the **OUTSIDE**.



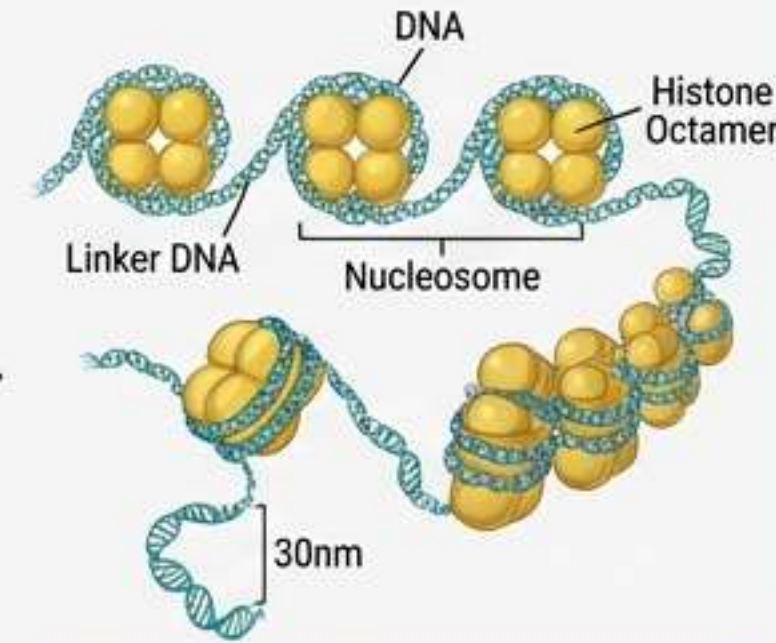
Mnemonic:
Bases are Shy (Inside).
Phosphates Fly (Outside).

Question 53 | The Nucleosome

The Nucleosome:

- A. Is the basic unit of a chromosome.**
- B. Consists of DNA, RNA, and protein.
- C. DNA makes one single turn.
- D. Specific to prokaryotes.
- E. Histones are rich in acidic amino acids.

Teacher Mode: The **Nucleosome** is the packing unit of Eukaryotic chromatin. DNA wraps ~1.65 times around **8 Histone proteins**. Histones are **Basic (+)** to attract **Acidic (-)** DNA.



Mnemonic: Opposites Attract:
(+) Histones hug (-) DNA.

Concept Review | RNA Types

Review: What are the 3 major RNA types and their abundances?

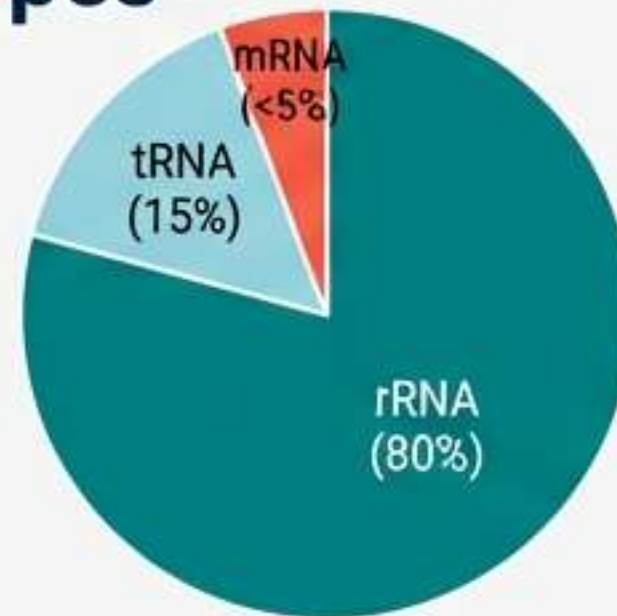
1. **rRNA** (Ribosomal)
2. **tRNA** (Transfer)
3. **mRNA** (Messenger)

Teacher Mode:

rRNA (80%): Structural factory (Ribosome).

tRNA (15%): Transports Amino Acids (Cloverleaf).

mRNA (<5%): Carries the code. Short lifespan.

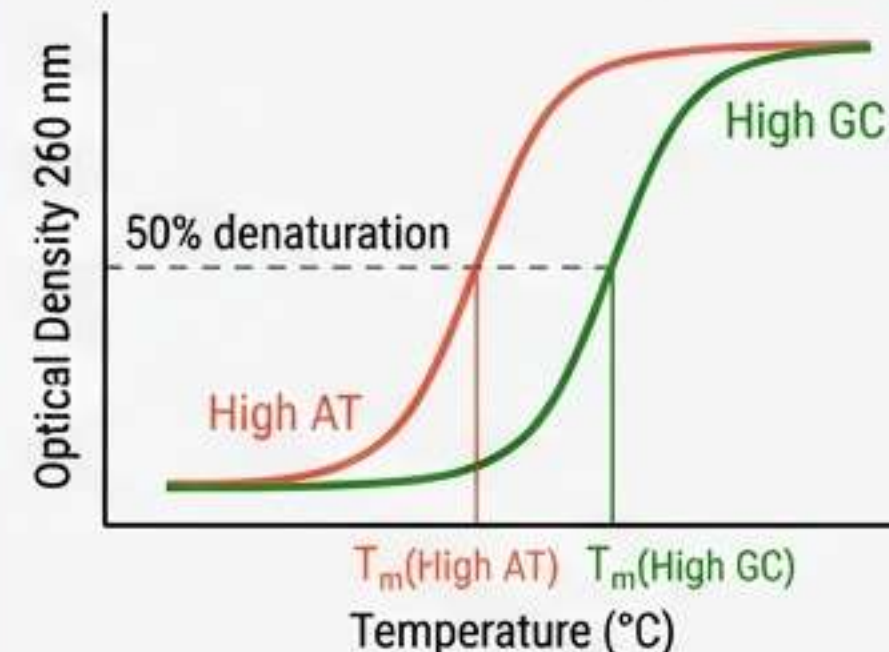


Mnemonic: rRNA is Really abundant (80%).
mRNA is Mini (<5%).

Concept Review | Melting Temp (T_m)

Review: What determines the T_m of DNA?

Answer: GC Content & Length.



Teacher Mode: T_m is the temp where 50% of DNA denatures.

GC Content: G-C pairs have 3 Hydrogen Bonds (harder to break than A-T's 2 bonds).

Length: Longer molecules = more bonds = Higher T_m .

Mnemonic:
G-C = Greater Connection
(3 bonds = Higher Heat)